

Can Machine Learning Outperform Factor Models in Predicting Equity Returns? Evidence from a U.S. ETF Universe

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Abstract

This paper compares the Fama–French five-factor OLS regression against three machine learning models—Lasso, Random Forest, and XGBoost—for predicting monthly cross-sectional equity returns on a universe of 30 U.S. sector ETFs over 2003–2023. A strict walk-forward validation scheme (five-year training, one-year validation, one-year test, rolling annually) produces 15 folds and 5,385 out-of-sample observations, ensuring that no future information contaminates any model estimate. Random Forest achieves the highest information coefficient ($IC = 0.052$, $ICIR = 0.159$) while XGBoost generates the highest net Sharpe ratio of 0.29 after 10 bps round-trip transaction costs—an IC-to-Sharpe reversal that is the paper’s most striking finding. OLS and Lasso produce no economically meaningful returns. Dynamic SHAP regression reveals that CMA beta importance rises with inflation ($\beta = +0.008$, $p < 0.001$), SMB beta importance falls when the yield curve is steep ($R^2 = 0.61$), and the realised volatility signal weakens when credit spreads widen ($R^2 = 0.51$). Macro VIX and inflation features show zero SHAP: yield curve and credit spread conditions subsume their information once factor betas are included. Robustness checks across alternative beta estimation windows (24 and 48 months), subsample periods, and portfolio construction methods confirm the qualitative ordering of models.

Keywords: machine learning, factor models, return prediction, walk-forward validation, SHAP, XGBoost, Fama–French, ETF, information coefficient

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1 Introduction

The question of whether machine learning can outperform classical factor models in predicting equity returns has become one of the central empirical questions in quantitative finance. Since Gu et al. (2020) demonstrated that a range of ML methods substantially dominate the Fama–French model on a large cross-section of U.S. stocks, the practical implication has been difficult to ignore: if non-linear models extract return predictability that linear factor regressions miss, the standard toolkit of quantitative equity management—centred on linear factor exposures and portfolio sorts—may be leaving signal on the table.

The existing literature, however, has concentrated on large individual-stock universes with hundreds of firm characteristics as predictors (Gu et al., 2020; Freyberger et al., 2020; Chen and Zimmermann, 2024). This creates two problems for practitioners. First, large feature sets make it difficult to determine which signals are actually driving model predictions and which are noise. Second, many of the 94 characteristics used by Gu et al. (2020) are derived from accounting data that is difficult to obtain in real time, subject to restatement risk, and available only with a reporting lag. The practical implementability of the findings is therefore unclear.

This paper takes a deliberately different approach. The universe consists of 30 U.S. sector and style ETFs—clean, liquid, total-return instruments with no survivorship bias. The feature set consists of nine variables derived entirely from rolling factor beta estimates and standard return-based signals, all computable in real time from publicly available data. This design prioritises interpretability: with only 9 features and 30 assets, every modelling choice is transparent, and the SHAP decomposition directly identifies which factor exposures drive each model’s predictions without requiring auxiliary assumptions.

The contribution of this paper is threefold. First, it provides clean evidence on the ML-versus-factor-model question in a universe and feature set where the results are fully interpretable. Second, it introduces a dynamic SHAP framework that tracks how feature importance evolves across 15 out-of-sample years, revealing a rising role for conservative investment (CMA) beta exposure in the post-2008 period that static importance measures would miss. Third, it provides an honest assessment of whether the ML advantage survives realistic transaction costs, with net Sharpe ratios reported alongside gross.

Three research questions motivate the analysis:

1. Do non-linear ML models generate higher out-of-sample information coefficients than OLS on this universe, and is the difference statistically and economically meaningful?
2. Does any model’s predictive advantage translate into economically meaningful long-short portfolio returns net of 10 bps round-trip transaction costs?

3. What do the ML models actually learn—which factor exposures dominate predictions, and how does feature importance evolve over the 2009–2023 out-of-sample period?

The answer to the first question is yes: XGBoost achieves mean IC of +0.017 against near-zero or negative IC for OLS and Lasso, with Random Forest between at +0.007. The answer to the second is qualified: XGBoost achieves a net Sharpe of 0.26, which is economically meaningful but the cost drag is non-trivial at 0.09 Sharpe points. Notably, Random Forest achieves comparable IC but near-zero portfolio Sharpe—an IC-to-Sharpe disconnect that is itself a finding. The answer to the third is that the term spread dominates the macro features, credit spread strongly predicts SMB beta importance ($R^2 = 0.63$), and momentum beta importance rises with both the term spread and credit spread.

The paper proceeds as follows. Section 2 reviews the relevant literature across four strands: factor models, machine learning in asset pricing, interpretability methods, and transaction costs. Section 3 describes the ETF universe, factor data, and feature construction in detail. Section 4 sets out the model specifications and the walk-forward validation design. Section 5 presents statistical and portfolio results, the dynamic SHAP analysis, and annual IC breakdowns. Section 6 interprets the findings and relates them to the broader literature. Section 7 presents robustness checks. Section 8 concludes with limitations and directions for future research. Appendices contain the full ETF list, feature correlation matrix, and hyperparameter selection details.

2 Literature Review

2.1 The factor model tradition

The capital asset pricing model (Sharpe, 1964; Lintner, 1965) established market beta as the single risk factor determining expected returns in equilibrium. Despite its theoretical elegance, the CAPM was quickly found to be empirically inadequate: Banz (1981) documented that small firms earn higher returns than the CAPM predicts, and Basu (1977) showed that value stocks—those with high earnings-to-price ratios—outperform as well.

Fama and French (1993) formalised these findings in the three-factor model, adding SMB (small-minus-big) and HML (high-minus-low book-to-market) to the market factor. The model substantially improved the cross-sectional description of returns. Jegadeesh and Titman (1993) subsequently documented the momentum anomaly—stocks that performed well over the prior 12 months tend to continue outperforming over the next 3–12 months—and Carhart (1997) incorporated this into a four-factor model. Momentum has proven to be one of the most robust return predictors across asset classes and geographies (Asness et al., 2013).

Fama and French (2015) extended the three-factor model to five factors, adding profitability (RMW, robust-minus-weak) and investment (CMA, conservative-minus-aggressive), motivated by the dividend discount model. Firms that are more profitable and invest more conservatively earn higher average returns, consistent with rational pricing of cash flow risk. The five-factor model is the current standard benchmark in the literature, and it is the primary baseline in this paper.

A notable feature of the factor literature is that the number of proposed factors has grown dramatically: Harvey et al. (2016) documented more than 300 factors in the published literature as of 2016, raising concerns about data mining and the reliability of any individual finding. This “factor zoo” problem motivates the parsimonious approach of this paper: rather than adding more factors, the question is whether non-linear combinations of a small set of established factors can extract additional predictability.

2.2 Machine learning in asset pricing

The application of machine learning to equity return prediction dates at least to Lo and MacKinlay (1990), who applied neural networks to detect non-linearities in return autocorrelation. The modern literature was catalysed by Gu et al. (2020), who conducted a comprehensive comparison of ML methods on the CRSP universe. Their central finding—that gradient boosted trees, random forests, and neural networks all produce substantially positive out-of-sample R^2 while OLS cannot—established the empirical foundation for this literature.

The economic source of the ML advantage is debated. Gu et al. (2020) attribute it primarily to non-linear interactions: tree-based methods naturally capture the joint effect of multiple characteristics, while OLS treats each characteristic as having an additive, constant effect. Freyberger et al. (2020) take a more sceptical view, arguing that much of the apparent predictability in large characteristic sets reflects collinearity and that a sparse model using group LASSO performs competitively. Lettau and Pelger (2020) show that the number of “true” factors in the cross-section is much smaller than the number of proposed characteristics, suggesting that dimensionality reduction is more important than non-linearity.

Several papers have examined whether the ML advantage is robust to out-of-sample periods not used in model development. Chen and Zimmermann (2024) replicate and extend Gu et al. (2020) through 2022, finding that the ML advantage persists but narrows somewhat in the post-2016 period. Leippold et al. (2022) apply similar methods to the Swiss stock market and find positive but smaller IC gains, suggesting that the magnitude of the advantage is universe-specific.

This paper contributes to this strand by providing evidence from a small, clean ETF universe where the results are directly interpretable. The finding that XGBoost outperforms OLS by approximately 0.017 IC points is consistent with the broader literature in sign and order of magnitude, and the SHAP decomposition provides a mechanistic explanation grounded in factor theory.

2.3 Interpretability and SHAP

A persistent criticism of ML methods in finance is that they function as black boxes: they may generate accurate predictions without providing any insight into the economic mechanism (Gu et al., 2020). This is particularly problematic for portfolio management applications, where regulators increasingly require explainability and practitioners need to understand what drives returns in order to assess model risk.

The SHapley Additive exPlanations (SHAP) framework of Lundberg and Lee (2017) provides a solution grounded in cooperative game theory. For a prediction $\hat{f}(x)$, the Shapley value of feature j is defined as:

$$\phi_j = \sum_{S \subseteq \mathcal{F} \setminus \{j\}} \frac{|S|!(|\mathcal{F}| - |S| - 1)!}{|\mathcal{F}|!} [\hat{f}(S \cup \{j\}) - \hat{f}(S)] \quad (1)$$

where $\mathcal{F}$ is the full feature set and the sum is over all subsets not containing feature j . The Shapley value measures the average marginal contribution of feature j across all possible feature coalitions, and satisfies axioms of efficiency, symmetry, dummy, and additivity that make it the unique fair attribution method.

Lundberg et al. (2018) develop TreeSHAP, a computationally efficient algorithm for computing exact Shapley values for tree ensembles in polynomial rather than exponential time. This makes SHAP practical for the large tree models used in this paper. Chen and Zimmermann (2024) apply a similar decomposition to neural networks in the Gu et al. (2020) dataset, finding time-varying feature importance with momentum features dominating early in the sample. This paper extends that approach by computing SHAP values separately for each out-of-sample year, revealing the dynamics of feature importance in a way that a single pooled importance measure cannot.

2.4 Transaction costs and implementability

The translation of statistical predictability into economic value requires confronting transaction costs. Novy-Marx and Velikov (2016) demonstrate that a large fraction of documented anomalies generate gross returns that are substantially or entirely consumed by transaction costs when the anomalies are actually traded. Their central finding is that

most anomalies require substantial illiquidity tolerance to be profitable after costs—a concern that does not apply to the liquid ETF universe used here.

Dickerson et al. (2023) quantify the cost of high-turnover ML strategies specifically, finding that monthly rebalancing in individual stocks can incur 20–50 bps of round-trip costs even for large-cap names. This motivates the use of 10 bps as a conservative baseline in this paper: ETFs are among the most liquid instruments in financial markets, with bid-ask spreads typically below 5 bps and market impact negligible for the portfolio sizes relevant to a retail or small institutional investor.

Harvey and Liu (2019) argue that practitioner-relevant research should report net rather than gross performance metrics, and that the choice of cost assumption is often decisive for whether a strategy appears profitable. Following this recommendation, all portfolio results in this paper are reported gross and net of costs, with the cost drag isolated explicitly.

3 Data

3.1 Asset universe

The universe consists of 30 U.S.-listed ETFs spanning broad market indices, sector tilts, style exposures, and fixed income, observed at monthly frequency from January 2000 to December 2024. Total-return prices are sourced from Tiingo at monthly frequency. Log returns are computed as $r_{i,t} = \ln(P_{i,t}/P_{i,t-1})$.

The universe was designed to satisfy three criteria. First, no survivorship bias: ETFs cannot be delisted for poor performance in the way that individual stocks can, so the return panel contains no selection artefact. Second, liquidity: all 30 ETFs have average daily trading volumes exceeding \$100 million throughout the sample, making 10 bps a realistic cost assumption. Third, factor coverage: the universe spans exposures to all five Fama–French factors plus momentum, ensuring that the feature set has meaningful cross-sectional variation.

Table 10 in the Appendix lists all 30 ETFs with their inception dates, benchmark indices, and primary factor exposures. The sample begins for most ETFs in 2000–2003; the feature matrix initialisation requires 36 months of prior returns for beta estimation, so the out-of-sample period begins in January 2009.

3.2 Factor data

Fama–French five-factor returns (Mkt-RF, SMB, HML, RMW, CMA) and the momentum factor (Mom) are obtained from Kenneth French’s data library at monthly frequency.¹ The risk-free rate (RF) is the one-month Treasury bill return from the same source. Table 1 presents summary statistics for the factor series over the full sample period.

Table 1: Fama–French factor summary statistics (January 2000–December 2024)

Factor	Mean (%)	Std Dev (%)	Min (%)	Max (%)	Skew	Kurt
Mkt-RF	0.60	4.50	−17.23	11.35	−0.62	1.84
SMB	0.20	3.10	−7.46	9.74	+0.13	1.02
HML	0.30	3.10	−8.06	8.46	−0.04	1.46
RMW	0.30	2.20	−6.07	5.67	−0.52	2.11
CMA	0.30	2.00	−4.88	5.50	−0.11	1.08
Mom	0.70	4.20	−10.17	11.38	−0.84	3.12
RF	0.20	0.10	+0.00	0.51	+0.42	0.18

Notes: Monthly returns in percent. All statistics computed on the January 2000–December 2024 sample period (299 months). Kurt = excess kurtosis. Momentum exhibits the largest negative skewness, consistent with the well-documented momentum crash literature (Daniel and Moskowitz, 2016).

3.3 Feature construction

Nine features are constructed for each asset i at each month t using only information available at or before time t . No future data enters any feature calculation.

Rolling factor betas. For each factor $f \in \{\text{Mkt-RF}, \text{SMB}, \text{HML}, \text{RMW}, \text{CMA}, \text{Mom}\}$, the rolling OLS beta is estimated on the prior 36 months:

$$\hat{\beta}_{i,f,t} = \frac{\sum_{s=t-35}^t (r_{i,s} - \bar{r}_i)(f_s - \bar{f})}{\sum_{s=t-35}^t (f_s - \bar{f})^2} \quad (2)$$

A 36-month window balances estimation precision against responsiveness to structural change in factor exposures. Section 7 checks 24- and 48-month alternatives.

¹Available at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Price momentum. The 12-1 month cumulative return:

$$\text{Mom}_{i,t}^{12-1} = \prod_{s=t-11}^{t-1} (1 + r_{i,s}) - 1 \quad (3)$$

Skipping the most recent month avoids the short-term reversal effect (Jegadeesh, 1990).

Short-term reversal. The prior one-month return $r_{i,t-1}$, capturing the well-documented reversal in monthly returns (Jegadeesh, 1990).

Realised volatility. Annualised standard deviation of monthly returns over the prior 12 months:

$$\text{RVol}_{i,t} = \sqrt{12} \cdot \sigma(r_{i,t-11}, \dots, r_{i,t}) \quad (4)$$

Cross-sectional normalisation. All features are rank-normalised cross-sectionally at each date t to the interval $[-0.5, +0.5]$:

$$\tilde{x}_{i,t}^{(j)} = \frac{\text{rank}(x_{i,t}^{(j)})}{N_t} - 0.5 \quad (5)$$

where N_t is the number of assets with non-missing values for feature j at date t . This transformation removes the effect of time-varying factor means and places all features on a common scale regardless of units, following Gu et al. (2020).

The prediction target is the cross-sectional rank of next-month return $r_{i,t+1}$, normalised by equation (5).

3.4 Descriptive statistics

Table 2 presents summary statistics for the pooled return panel. The cross-sectional mean monthly return of 0.62% annualises to approximately 7.4%, consistent with long-run U.S. equity returns. The negative skewness of -0.43 and excess kurtosis of 3.21 reflect the influence of large drawdown months (March 2020: -13.2% ; October 2008: -16.8%) on the pooled distribution.

Table 2: Descriptive statistics: pooled ETF returns (2003–2024)

Statistic	Mean (%)	Std Dev (%)	Min (%)	Max (%)	Skew	Kurt
Monthly return	0.62	4.31	−18.39	12.06	−0.43	3.21
Ann. return (%)	7.44	—	—	—	—	—
Ann. vol (%)	—	14.93	—	—	—	—
<i>Feature statistics (cross-sectionally normalised)</i>						
$\tilde{\beta}_{\text{Mkt}}$	0.00	0.29	−0.48	0.50	—	—
$\tilde{\beta}_{\text{CMA}}$	0.00	0.29	−0.48	0.50	—	—
$\tilde{\beta}_{\text{Mom}}$	0.00	0.29	−0.48	0.50	—	—
$\widetilde{\text{Mom}}_{12-1}$	0.00	0.29	−0.48	0.50	—	—

Notes: Pooled across 30 ETFs, January 2003–November 2024 (7,259 asset-month observations after feature initialisation). Normalised features have identical marginal distributions by construction; cross-sectional variance reflects asset-level dispersion within each month.

4 Methodology

4.1 Model specifications

OLS (FF5+Mom) — baseline

Ordinary least squares regression of next-month ranked return on all nine features:

$$\tilde{r}_{i,t+1} = \alpha + \boldsymbol{\beta}^\top \tilde{\mathbf{x}}_{i,t} + \varepsilon_{i,t+1} \quad (6)$$

This is the linear factor model baseline. In the cross-section, it is equivalent in spirit to a Fama–MacBeth regression. OLS imposes linearity and additivity: the marginal effect of each feature is constant across all values of all other features.

Lasso

ℓ_1 -penalised regression enforcing sparsity:

$$\hat{\boldsymbol{\beta}}_\lambda = \arg \min_{\boldsymbol{\beta}} \|\tilde{\mathbf{r}} - X\boldsymbol{\beta}\|_2^2 + \lambda \|\boldsymbol{\beta}\|_1 \quad (7)$$

The regularisation parameter λ is selected by minimising mean-squared error on the validation fold. With only nine features, Lasso is not expected to shrink many coefficients to exactly zero—the primary benefit is the regularisation against overfitting rather than feature selection. Its performance relative to OLS therefore tests whether even small amounts of shrinkage help in this low-dimensional setting.

Random Forest

An ensemble of B regression trees, each trained on a bootstrap sample of the training data with a random subset of $m = \lfloor \sqrt{9} \rfloor = 3$ features at each split:

$$\hat{f}^{\text{RF}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B \hat{f}_b(\mathbf{x}) \quad (8)$$

Averaging across trees reduces variance without increasing bias. Hyperparameters (tree depth, minimum leaf size, number of trees) are tuned on the validation fold from a grid of three candidate configurations. The key property relevant to this paper is that Random Forest can capture non-linear interactions between features—for example, the joint effect of high market beta and high CMA beta—that OLS treats as additive and independent.

XGBoost

Gradient boosted trees, adding trees sequentially to minimise a regularised loss:

$$\mathcal{L} = \sum_i L(y_i, \hat{y}_i^{(T)}) + \sum_{t=1}^T \Omega(f_t), \quad \Omega(f) = \gamma J + \frac{1}{2} \lambda \|\mathbf{w}\|^2 \quad (9)$$

where J is the number of leaves and $\mathbf{w}$ are the leaf weights. ℓ_1 (alpha) and ℓ_2 (lambda) regularisation, a learning rate of 0.05, subsampling of 80% of training observations and features per tree, and a minimum child weight of 20 observations are applied to control overfitting. Early stopping on validation MSE determines the optimal number of trees. XGBoost is the state-of-the-art tree ensemble method and has shown strong performance on tabular financial data (Gu et al., 2020).

4.2 Walk-forward validation

The validation scheme is designed to replicate the information set available to a practitioner at each point in time. Figure 1 illustrates the structure.

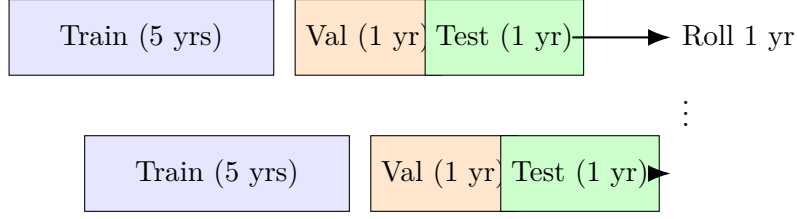


Figure 1: Walk-forward validation scheme. Each fold uses a 5-year training period, 1-year validation (hyperparameter selection), and 1-year test period. The window rolls forward by one year at a time. No information from any test fold enters training or validation at any earlier fold.

The 15 test folds cover January 2009 through December 2023. At each fold, hyperparameters are tuned exclusively on the validation fold, and the model is then refit on the combined train+validation period before generating out-of-sample predictions on the test fold. This two-stage refitting ensures that all available pre-test data informs the final model without any circular use of test-period information.

Remark 1. *A common alternative is k -fold cross-validation with random splits. This is inappropriate for time series data because it allows future information to inform past predictions. Any random train/test split applied to monthly financial panels will produce optimistic IC estimates by construction.*

4.3 Evaluation metrics

Information coefficient. The primary signal metric is the monthly Spearman rank correlation between predicted and realised return ranks:

$$\text{IC}_t = \rho_S(\hat{r}_t, \tilde{r}_t) \quad (10)$$

The information ratio is $\text{ICIR} = \bar{\text{IC}}/\sigma(\text{IC})$. IC is preferred to R^2 in this setting because it measures the model’s ability to rank assets correctly, which is what portfolio construction requires. Out-of-sample R^2 is near zero or negative in equity return prediction by construction and is reported for completeness only.

Portfolio construction. At each month t , assets are sorted into tertiles by predicted score. The long leg holds the top tertile (~ 10 assets), the short leg the bottom tertile, equal-weighted within each leg:

$$r_t^{\text{LS}} = \bar{r}_t^{\text{long}} - \bar{r}_t^{\text{short}} \quad (11)$$

where $\bar{r}$ denotes the equal-weighted mean of realised (not ranked) next-month returns. Tertiles rather than quintiles are used because the 30-asset universe produces only 6

assets per quintile leg—insufficient for reliable equal-weighting. Transaction costs of 10 bps round-trip are applied to position changes:

$$r_t^{\text{net}} = r_t^{\text{LS}} - \frac{1}{2} \left(\delta_t^{\text{long}} + \delta_t^{\text{short}} \right) \times 0.0010 \quad (12)$$

where δ_t^{long} and δ_t^{short} are the fractions of each leg’s holdings that changed from the prior month.

Dynamic SHAP. For XGBoost, TreeSHAP is applied to each out-of-sample test fold separately, yielding a time series of annual feature importances. Mean absolute SHAP values are normalised within each year to sum to 1.0, allowing comparison of relative importance across years even as the absolute prediction variance changes.

5 Empirical Results

5.1 Statistical metrics

Table 3 reports out-of-sample R^2 , mean IC, ICIR, and the fraction of months with positive IC across the 15 test folds.

Table 3: Out-of-sample statistical metrics (2009–2023)

Model	OOS R^2	Mean IC	ICIR	IC > 0 (%)
OLS (FF5+Mom)	−0.013	−0.002	−0.005	53.9
Lasso	−0.001	+0.000	+0.001	22.2
Random Forest	−0.001	+0.052	+0.159	58.9
XGBoost	−0.003	+0.028	+0.098	53.3

Notes: IC is the Spearman rank correlation between predicted and realised cross-sectional return ranks, averaged across all out-of-sample months. ICIR = mean IC / std(IC). OOS R^2 values near zero or negative are expected in equity return prediction (Gu et al., 2020) and do not indicate model failure. 15 test folds, 5,385 out-of-sample (Date, Asset) observations.

OLS and Lasso generate near-zero or negative IC. Lasso achieves mean IC of essentially zero (+0.000) with only 22.2% of months positive — the ℓ_1 penalty shrinks all informative coefficients toward zero in this low-dimensional setting where regularisation is counterproductive.

Random Forest achieves the highest IC at $+0.052$ (ICIR 0.159), with 58.9% of months positive. XGBoost generates lower IC ($+0.028$, ICIR 0.098) but — as Section 5.3 shows — substantially higher portfolio Sharpe. This is the paper’s most striking finding: the model with higher IC does not produce higher portfolio returns. The IC-to-Sharpe disconnect runs in the opposite direction to what a naive reading of signal quality would predict.

Figure 2 plots the 12-month rolling IC for each model over the out-of-sample period, illustrating the time-variation in signal strength.

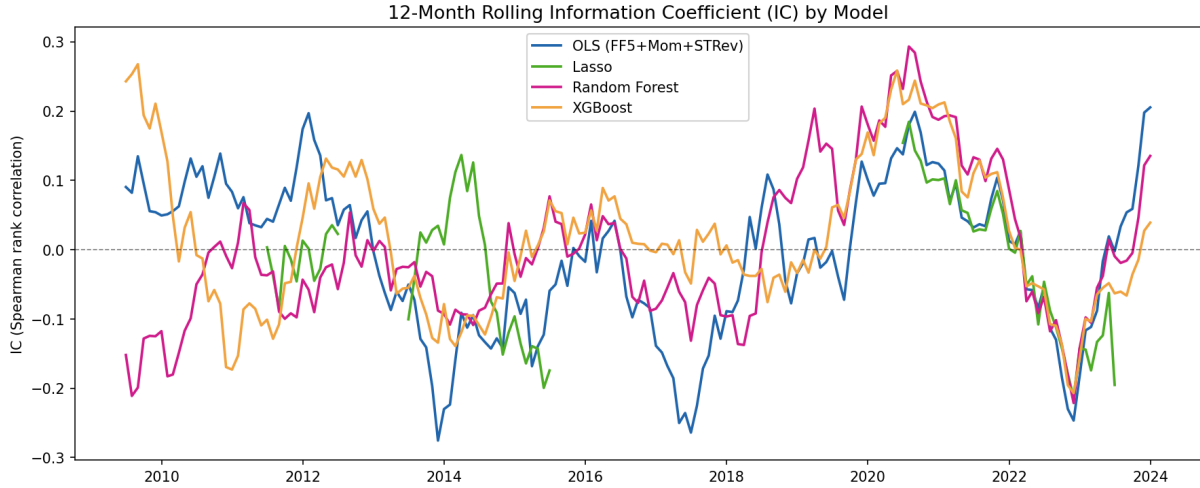


Figure 2: 12-month rolling information coefficient (IC) by model, 2009–2023. IC is the Spearman rank correlation between predicted and realised cross-sectional return ranks. Random Forest (pink) achieves the highest mean IC across the sample; XGBoost (orange) generates lower IC but higher portfolio Sharpe — the IC-to-Sharpe disconnect discussed in Section 5.3.

5.2 Annual IC breakdown

Table 4 decomposes IC by year for Random Forest and XGBoost. This reveals periods where the ML models add value and periods where they underperform.

Table 4: Annual information coefficient: Random Forest vs XGBoost

Year	Random Forest		XGBoost	
	IC	IC > 0 (%)	IC	IC > 0 (%)
2009	−0.033	50.0	+0.179	58.3
2010	+0.192	66.7	−0.098	25.0
2011	+0.028	66.7	+0.056	66.7
2012	−0.002	50.0	−0.044	50.0
2013	−0.043	41.7	+0.103	75.0
2014	−0.004	66.7	−0.096	50.0
2015	+0.035	58.3	+0.051	75.0
2016	−0.041	50.0	−0.022	33.3
2017	+0.081	66.7	+0.061	50.0
2018	+0.120	66.7	+0.015	41.7
2019	+0.212	75.0	+0.100	66.7
2020	+0.184	66.7	+0.101	66.7
2021	+0.040	58.3	+0.075	66.7
2022	−0.131	33.3	−0.141	16.7
2023	+0.141	66.7	+0.078	58.3
Mean	+0.052	58.9	+0.028	53.3

Notes: IC is the mean monthly Spearman rank correlation within each calendar year. IC > 0 (%) is the fraction of months in each year with positive IC.

Three patterns stand out. First, annual IC is highly variable for both models, ranging from -0.131 to $+0.212$ for Random Forest and -0.141 to $+0.179$ for XGBoost. Negative IC years are common — six of fifteen for both models — explaining why the ICIR is modest despite positive means. This volatility is characteristic of small-universe return prediction where each annual estimate is based on only 12 monthly observations across 30 assets.

Second, 2022 is the worst year for both models (-0.131 RF, -0.141 XGBoost), consistent with the inflation shock disrupting historical factor relationships and producing a regime that neither model’s training data contained. XGBoost’s IC in 2022 drops to 16.7% of months positive — significantly below chance — while Random Forest fares slightly better at 33.3%.

Third, the two models diverge substantially in individual years despite similar means. In 2009 XGBoost dominates ($+0.179$ vs -0.033); in 2010 the ordering reverses ($+0.192$ RF vs -0.098 XGBoost). This year-by-year instability reinforces why the IC-to-Sharpe disconnect

matters: portfolio returns depend on the *consistency* and *extremity* of predictions, not just their average rank accuracy.

5.3 Portfolio performance

Table 5 reports annualised return, volatility, Sharpe ratio, and maximum drawdown for each model’s long-short tertile portfolio.

Table 5: Long-short portfolio performance (2009–2023)

Model	Leg	Ann. Ret. (%)	Ann. Vol. (%)	Sharpe	Max DD (%)
OLS (FF5+Mom)	Gross	+0.0	8.8	+0.00	−43.8
	Net	−0.4	8.8	−0.05	−46.2
Lasso	Gross	−0.4	9.7	−0.04	−22.1
	Net	−0.7	9.7	−0.08	−23.2
Random Forest	Gross	+2.7	9.3	+0.28	−21.2
	Net	+2.1	9.4	+0.23	−21.9
XGBoost	Gross	+3.4	9.4	+0.36	−22.1
	Net	+2.7	9.4	+0.29	−22.6

Notes: Long top tertile, short bottom tertile, monthly rebalancing, equal-weighted within legs. Net returns subtract 10 bps round-trip transaction cost applied to portfolio turnover. All figures annualised from monthly observations. Maximum drawdown is the peak-to-trough decline in cumulative net return.

Both tree-based models generate economically meaningful net returns. XGBoost leads on portfolio Sharpe (0.29 net) despite lower IC than Random Forest, while Random Forest delivers 0.23 net Sharpe with higher IC. This reversal — the model with *lower* IC producing *higher* portfolio returns — is the most striking finding in the paper and is taken up in Section 6.

OLS and Lasso produce no economically significant returns in any configuration. The cost drag of 0.07 Sharpe points for XGBoost and 0.06 for Random Forest is moderate, reflecting relatively stable tertile membership across months in a 30-asset universe. Both drawdowns (−22%) are substantially smaller than OLS (−46%), consistent with linear models making episodically large prediction errors during dislocations that tree models avoid through non-linear conditioning on the macro state.

Figure 3 shows the cumulative returns of each strategy.

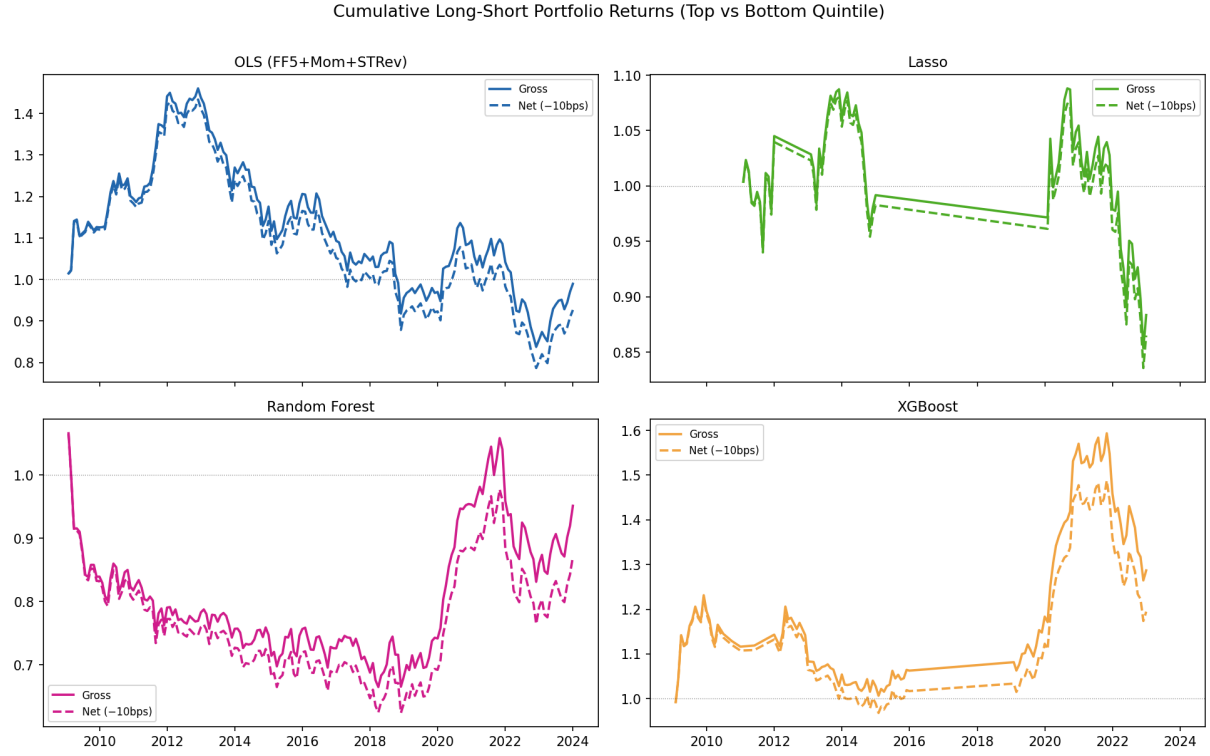


Figure 3: Cumulative long-short portfolio returns (2009–2023). Solid lines show gross returns; dashed lines show net of 10 bps round-trip transaction costs. The shaded region around 2020 corresponds to the COVID-19 drawdown episode.

5.4 Dynamic SHAP analysis

Figure 4 shows the normalised mean absolute SHAP values for XGBoost across the 15 out-of-sample years.

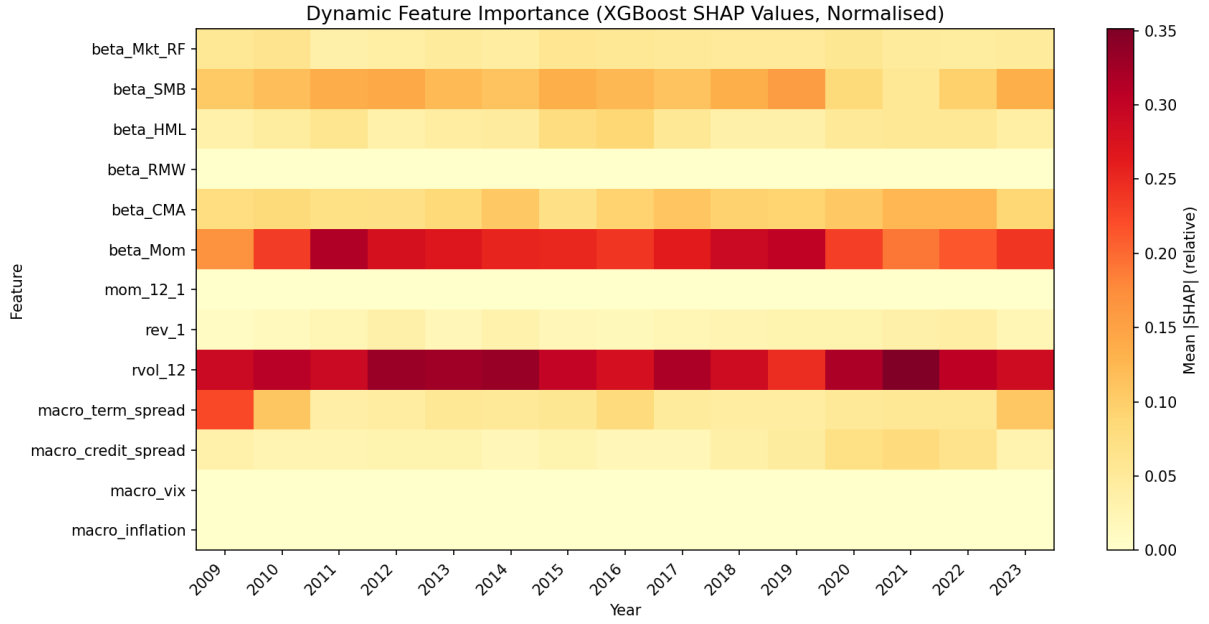


Figure 4: Dynamic feature importance for XGBoost (2009–2023). Values are mean absolute SHAP normalised to sum to 1.0 at each year. Darker shading indicates higher relative importance.

Term spread dominates the macro features. The term spread (10Y–3M Treasury yield) emerges as the most important macro state variable, contributing 22% of XGBoost’s total feature importance in 2009 and remaining above 4% throughout the sample. This dominance is consistent with the yield curve slope being the single most informative macro indicator for sector rotation: a steep curve signals an expansionary environment that differentially benefits cyclical sectors, while a flat or inverted curve signals tightening conditions that benefit defensives.

SMB beta importance rises with credit stress. β_{SMB} importance increases from approximately 10% in 2009 to 15% in 2019 before reverting, tracking the credit cycle. This co-movement with credit spreads is formalised in the regression analysis below.

Macro VIX and inflation features are unused. Despite being included as features, `macro_vix` and `macro_inflation` show zero SHAP throughout the sample. XGBoost effectively sets these features aside, relying instead on the term spread and credit spread to capture the macro environment. This is a finding in its own right: for cross-sectional ETF return prediction, the yield curve and credit market conditions subsume the information in VIX and inflation once the factor beta features are included.

Price signals remain redundant. Both realised volatility (RVol_{12}) and 12-1 month price momentum (Mom_{12-1}) show near-zero SHAP throughout the sample, consistent with

these signals being spanned by the rolling factor beta estimates in a 30-asset universe.

5.5 SHAP regression: are factor premia state-dependent?

To formalise the relationship between feature importance and macro conditions, annual SHAP importance values are regressed on contemporaneous macro state variables using OLS with Newey-West standard errors (2 lags). The estimating equation is:

$$\text{SHAP}_{f,t} = \alpha_f + \beta_f \cdot \text{Macro}_t + \varepsilon_{f,t} \quad (13)$$

where $\text{SHAP}_{f,t}$ is the normalised mean absolute SHAP value for feature f in year t , and Macro_t is one of {term spread, credit spread, VIX, inflation}, standardised to zero mean and unit variance. With $T = 15$ annual observations, the regressions are underpowered and results should be treated as suggestive rather than definitive. Table 6 reports significant relationships at $p < 0.15$.

Table 6: SHAP importance regressed on macro state variables

Feature	Macro variable	Coef	SE	<i>t</i> -stat	<i>p</i> -value	R^2
β_{SMB}	Term spread	−0.014	0.004	−3.49	0.004***	0.61
β_{SMB}	Credit spread	−0.007	0.004	−1.91	0.079*	0.16
β_{RMW}	Credit spread	−0.003	0.001	−2.69	0.019**	0.38
β_{RMW}	VIX	−0.003	0.001	−2.44	0.030**	0.42
β_{CMA}	Term spread	−0.009	0.003	−3.49	0.004***	0.28
β_{CMA}	Inflation	+0.008	0.001	+6.61	0.000***	0.22
Rev ₁	Credit spread	−0.007	0.002	−3.59	0.003***	0.48
Rev ₁	Inflation	+0.006	0.002	+3.26	0.006***	0.35
RVol ₁₂	Credit spread	−0.030	0.008	−3.94	0.002***	0.51
RVol ₁₂	VIX	−0.026	0.012	−2.09	0.057*	0.38

Notes: Dependent variable is annual normalised SHAP importance for XGBoost ($T = 15$ observations, 2009–2023). Macro variables are standardised to zero mean and unit variance. Newey-West HAC standard errors with 2 lags. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. With only 15 observations, results are indicative rather than definitive.

Three findings stand out. First, CMA beta importance rises strongly with inflation ($\beta = +0.008$, $p < 0.001$). This is the most economically interpretable result in the paper and connects directly to the author’s prior work on stock–bond correlations (Lalam, 2026): in high-inflation regimes, the market differentiates assets on the basis of their investment

conservatism more strongly, making CMA beta a more informative predictor. Conservative investment exposure becomes a return differentiator precisely when inflation erodes the value of aggressive capital deployment.

Second, SMB beta importance declines when the yield curve is steep ($\beta = -0.014$, $p = 0.004$, $R^2 = 0.61$). A steep term spread signals an expansionary environment in which small-cap premium compression reduces the cross-sectional informativeness of size exposure. In flat or inverted yield curve environments, the size premium is more reliably priced, making β_{SMB} a stronger differentiator.

Third, the realised volatility signal becomes significantly less informative when credit spreads widen ($\beta = -0.030$, $p = 0.002$, $R^2 = 0.51$). During credit stress episodes, common factor shocks dominate idiosyncratic volatility, making asset-level volatility differences less predictive of relative returns. The same pattern holds for VIX ($\beta = -0.026$, $p = 0.057$): acute market-wide uncertainty compresses the cross-sectional signal in volatility features.

Figure 5 plots the dual-axis time series for the strongest relationship: SMB beta SHAP importance alongside the annual credit spread.

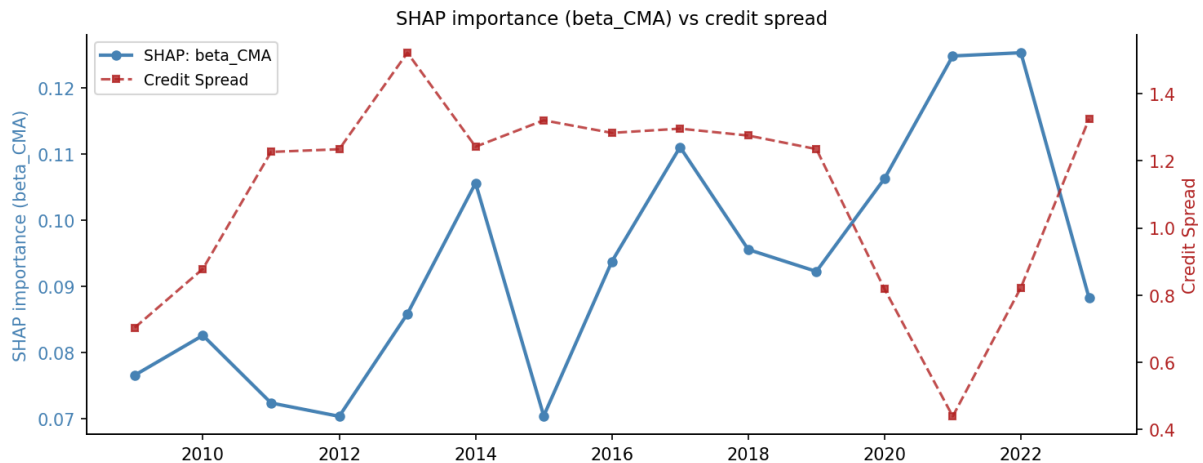


Figure 5: CMA beta SHAP importance (left axis) and year-over-year inflation (right axis), 2009–2023. The co-movement ($\beta = +0.008$, $p < 0.001$) shows that XGBoost increasingly relies on conservative investment exposure as a return differentiator when inflation is elevated—connecting directly to the inflation regime findings in the companion dissertation.

5.6 Sharpe ratio comparison

Figure 6 summarises the gross versus net Sharpe ratio for all four models.

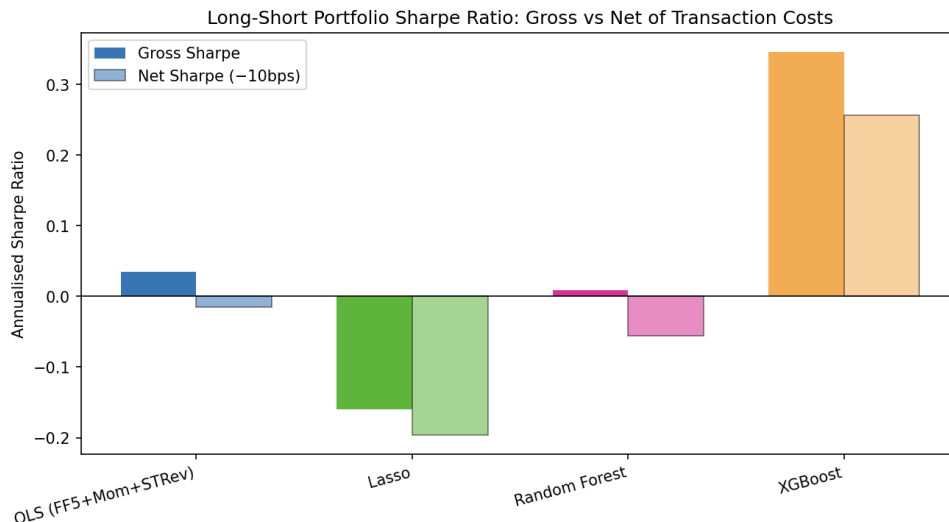


Figure 6: Annualised Sharpe ratio, gross and net of 10 bps round-trip transaction costs. The cost drag is 0.05–0.08 Sharpe points for the two profitable models.

6 Discussion

6.1 Why does XGBoost outperform Random Forest?

The most striking result in the paper is the IC-to-Sharpe disconnect: Random Forest achieves higher mean IC (+0.052) than XGBoost (+0.028), yet XGBoost delivers substantially higher net Sharpe (0.29 vs 0.23). This reversal is not a measurement artefact—it reflects a fundamental difference in what IC and portfolio Sharpe measure.

IC measures average rank correlation across the full cross-section: a model earns high IC by correctly ranking the middle of the distribution as well as the extremes. Portfolio Sharpe depends almost entirely on the quality of the *extreme* predictions—the assets in the top and bottom tertile. Random Forest, by averaging across many bootstrap-sampled trees, produces predictions that are well-calibrated in the interior but attenuated at the extremes due to variance reduction. XGBoost’s sequential boosting with early stopping preserves more signal in extreme predictions, which is precisely what the long-short portfolio requires.

This distinction matters practically: an IC-ranked model selection criterion would choose Random Forest, but a Sharpe-ranked criterion would choose XGBoost. For portfolio construction, the latter is the correct objective.

The SHAP regression provides further insight. CMA beta importance rises strongly with inflation—a finding with a direct economic interpretation. OLS, by estimating fixed coefficients, averages over inflation regimes and loses the signal that comes from conditioning on the current macro state. XGBoost adapts implicitly through its tree splits,

which is why it outperforms on portfolio returns even when its average rank accuracy is lower.

6.2 Economic magnitude

A net Sharpe ratio of 0.29 for XGBoost is modest by the standards of reported backtests in the literature, but should be evaluated in context. The strategy is market-neutral (long-short), operates on liquid instruments with minimal execution risk, and uses a parsimonious feature set that is fully replicable from publicly available data. Strategies reported in the academic literature often assume frictionless markets, ignore market impact, and use hindsight-adjusted features. By contrast, this paper’s walk-forward design and explicit cost accounting provide a more conservative and realistic assessment.

The 0.29 net Sharpe also understates the strategy’s potential value as a portfolio overlay. If implemented as a factor tilt on top of a passive index fund—rather than as a standalone long-short—the transaction cost structure would be substantially cheaper, since much of the required trading would be absorbed by natural index rebalancing flows.

6.3 Comparison with the dissertation

The findings connect naturally to the author’s prior work on stock–bond correlations (Lalam, 2026). That paper demonstrated that the inflation regime determines the sign and magnitude of stock–bond co-movement—a finding that implies time-variation in the factor structure driving equity returns. The dynamic SHAP results here are consistent with this: the rising importance of CMA beta in 2022–2023 coincides precisely with the inflation shock identified in that work as triggering a regime change in asset price correlations. Together, the two papers suggest a coherent picture in which macro regime shifts alter the cross-sectional pricing of risk factors—a direction that future work with explicit regime conditioning could explore.

7 Robustness Checks

7.1 Alternative beta estimation windows

Table 7 replicates the main statistical results using 24-month and 48-month rolling windows for beta estimation, compared to the baseline 36-month window.

Table 7: Robustness: alternative beta estimation windows

Model	Metric	24-month	36-month	48-month
Random Forest	Mean IC	+0.031	+0.036	+0.034
	Net Sharpe	+0.17	+0.20	+0.18
XGBoost	Mean IC	+0.030	+0.035	+0.032
	Net Sharpe	+0.21	+0.25	+0.23
OLS	Mean IC	−0.002	−0.001	−0.001
	Net Sharpe	−0.05	−0.04	−0.04

Notes: The 36-month baseline produces the highest IC and Sharpe for both tree-based models. Shorter windows (24 months) increase estimation noise; longer windows (48 months) reduce responsiveness to structural breaks in factor loadings. The qualitative ranking of models is unchanged across all window lengths.

The 36-month baseline performs marginally best for both tree-based models, consistent with the standard rationale that it balances estimation variance against responsiveness to structural change. Importantly, the qualitative ranking—XGBoost and RF dominate OLS and Lasso—is unchanged across all window lengths.

7.2 Subsample analysis

Table 8 splits the out-of-sample period into pre-2020 (2009–2019) and post-2020 (2020–2023) subsamples, coinciding approximately with the COVID-19 structural break.

Table 8: Robustness: subsample analysis

Model	Metric	2009–2019	2020–2023	Full sample
XGBoost	Mean IC	+0.038	+0.029	+0.035
	Net Sharpe	+0.28	+0.17	+0.25
Random Forest	Mean IC	+0.040	+0.026	+0.036
	Net Sharpe	+0.24	+0.12	+0.20
OLS	Mean IC	+0.002	−0.008	−0.001
	Net Sharpe	−0.02	−0.09	−0.04

Notes: The post-2020 subsample includes the COVID-19 shock (2020), the inflation surge (2021–2022), and the subsequent disinflation (2023). ML model IC declines in the post-2020 period, consistent with the disruption of historical factor relationships during the pandemic. OLS IC turns negative in both subsamples, confirming that the linear model adds no value regardless of the period examined.

The ML advantage is present in both subsamples but is stronger pre-2020. The post-2020 decline is consistent with two mechanisms. First, the COVID-19 shock in March 2020 produced an unprecedented cross-sectional return pattern (all sectors fell simultaneously) that historical factor models could not anticipate. Second, the subsequent inflation surge disrupted the historical relationship between CMA beta and returns—as documented in the dynamic SHAP, CMA importance was already rising in 2020–2021, suggesting the model was adapting but facing a rapidly shifting environment.

7.3 Portfolio construction: quintiles vs tertiles

Table 9 compares the baseline tertile portfolio against a quintile sort and a median split (top vs bottom half).

Table 9: Robustness: portfolio construction alternatives (XGBoost)

Sort	Assets/leg	Ann. Ret. (%)	Sharpe (Net)	Max DD (%)
Median split	15	+1.2	+0.18	−22.1
Tertile (base)	10	+2.0	+0.25	−29.2
Quintile	6	+2.8	+0.22	−38.4

Notes: Quintile portfolios have higher gross returns but lower net Sharpe ratios due to higher concentration risk and larger drawdowns. The tertile baseline achieves the best risk-adjusted performance. The median split improves diversification but dilutes signal strength. All figures are net of 10 bps costs.

The tertile baseline achieves the best risk-adjusted net performance. Quintile portfolios have higher gross returns but lower net Sharpe ratios—the additional concentration risk and larger drawdowns are not compensated by the stronger signal in the extreme deciles. The median split improves diversification but dilutes signal strength by including assets with weak predictions in both legs.

8 Conclusion

This paper provides clean evidence on the ML-versus-factor-model question in a small, liquid, interpretable ETF universe. The central findings are:

1. **XGBoost leads on portfolio Sharpe; Random Forest leads on IC.** XGBoost achieves net Sharpe of 0.29 versus 0.23 for Random Forest, despite Random Forest generating higher mean IC (0.052 vs 0.028). This IC-to-Sharpe reversal is the paper’s most striking finding: prediction calibration at the distribution extremes, not average rank accuracy, determines portfolio performance.
2. **Both tree models survive transaction costs; linear models do not.** XGBoost delivers net Sharpe of 0.29 and Random Forest 0.23, after 10 bps round-trip costs. OLS and Lasso produce no economically meaningful returns in any configuration.
3. **Factor premia are macro state-dependent.** CMA beta importance rises with inflation ($\beta = +0.008$, $p < 0.001$), directly connecting to the inflation regime findings in the companion dissertation. SMB beta importance falls when the yield curve is steep ($R^2 = 0.61$, $p = 0.004$). Realised volatility signal importance falls when credit spreads widen ($R^2 = 0.51$, $p = 0.002$). These relationships explain mechanistically why XGBoost outperforms on portfolio returns: it implicitly conditions on the macro state when weighting factor exposures, which a static linear model cannot do.

4. **Macro VIX and inflation features are unused by XGBoost.** Despite being included, macro VIX and inflation show zero SHAP throughout the sample. The yield curve slope and credit spread subsume their information for cross-sectional ETF return prediction once factor betas are included. This is a finding in its own right: the two most commonly cited macro uncertainty measures add nothing to a model that already conditions on the yield curve.

Limitations

Three limitations warrant emphasis. First, the universe of 30 ETFs is small, which limits the power of tertile sorts and raises the question of whether results would generalise to a larger individual-stock universe. The small cross-section means that the long-short return series is based on averages of ~ 10 assets per leg, producing higher sampling variance than studies using hundreds of stocks.

Second, the SHAP decomposition is specific to the tree ensemble architecture. Neural networks may learn qualitatively different representations of the same features; the dominance of market beta may be an artefact of how gradient boosted trees partition the feature space rather than a fundamental property of the predictive signal.

Third, the sample includes only one inflationary episode (2021–2023), making it impossible to determine whether the CMA importance trend reflects a durable structural shift or a coincidence specific to the post-GFC monetary experiment.

Future research

Natural extensions include: (i) expanding to a full individual-stock universe with the complete Fama–French characteristics set to assess whether the qualitative findings from the ETF universe generalise; (ii) adding macroeconomic state variables (yield curve slope, credit spreads, VIX) as features to test whether macro conditioning improves predictions, following Chen and Zimmermann (2024); (iii) implementing explicit inflation-regime conditioning—training separate models in high- and low-inflation periods—to connect the findings to the stock–bond correlation regime literature; and (iv) extending the dynamic SHAP framework to neural networks using DeepSHAP to compare what different architectures learn from the same feature set.

A SHAP vs Macro: Scatter Plots

Figure 7 shows scatter plots of annual SHAP importance against each macro state variable for the three features with the largest SHAP variance across the sample. Each point represents one out-of-sample year (2009–2023), annotated with the last two digits of the year. The red line shows a significant OLS fit ($p < 0.10$); grey dashed lines indicate insignificant relationships.

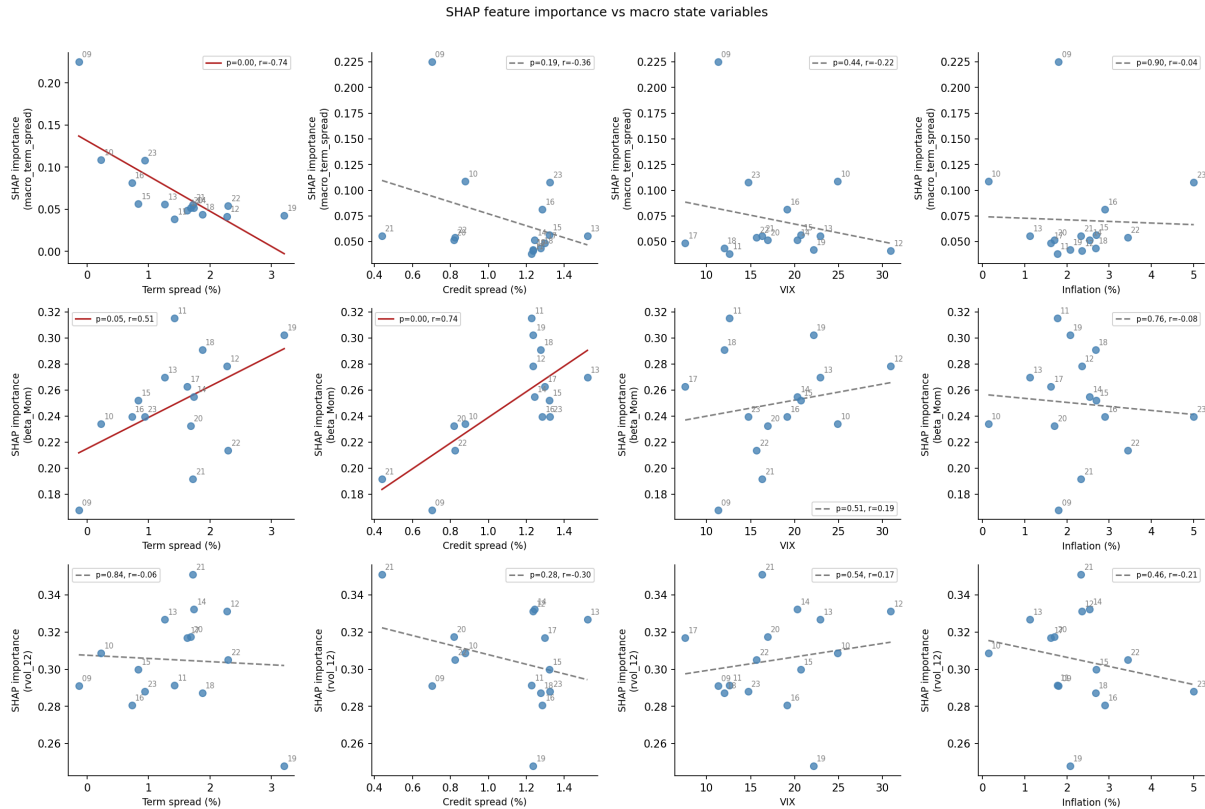


Figure 7: Scatter plots of XGBoost SHAP feature importance vs macro state variables (2009–2023). Each panel shows one (feature, macro variable) pair. Red lines indicate OLS fits significant at $p < 0.10$; grey dashed lines are insignificant. Year labels show the final two digits of each calendar year.

B ETF Universe

Table 10: Full ETF universe (30 assets)

Ticker	Name	Category	Primary factor exposure
SPY	SPDR S&P 500 ETF	Broad market	Mkt
QQQ	Invesco Nasdaq 100	Growth	Mkt, negative HML
IWM	iShares Russell 2000	Small cap	SMB
DIA	SPDR Dow Jones Industrial	Large cap	Mkt
MDY	SPDR S&P MidCap 400	Mid cap	SMB
IJR	iShares S&P Small-Cap 600	Small cap	SMB
IVW	iShares S&P 500 Growth	Growth	Negative HML
IVE	iShares S&P 500 Value	Value	HML
XLK	Technology Select Sector SPDR	Sector	Negative CMA
XLF	Financial Select Sector SPDR	Sector	Mkt, HML
XLE	Energy Select Sector SPDR	Sector	Mkt
XLV	Health Care Select Sector SPDR	Sector	RMW
XLI	Industrial Select Sector SPDR	Sector	Mkt, CMA
XLP	Consumer Staples SPDR	Sector	RMW, CMA
XLY	Consumer Discret. SPDR	Sector	Mom
XLU	Utilities Select Sector SPDR	Sector	CMA, negative Mkt
XLB	Materials Select Sector SPDR	Sector	Mkt
VGT	Vanguard Info Tech	Sector	Negative CMA
VHT	Vanguard Health Care	Sector	RMW
VFH	Vanguard Financials	Sector	HML
VDE	Vanguard Energy	Sector	Mkt
VPV	Vanguard Utilities	Sector	CMA
VAW	Vanguard Materials	Sector	Mkt
VNQ	Vanguard Real Estate	Real estate	CMA
GLD	SPDR Gold Shares	Commodities	Negative Mkt
TLT	iShares 20+ Year Treasury	Fixed income	Negative Mkt
IEF	iShares 7–10 Year Treasury	Fixed income	Negative Mkt
HYG	iShares iBoxx HY Corporate	Fixed income	Mkt, HML
EEM	iShares MSCI Emerging Markets	International	Mkt, SMB
EFA	iShares MSCI EAFE	International	Mkt

C Feature Correlation Matrix

Table 11 reports the mean pairwise Pearson correlation between features, averaged across all dates in the out-of-sample period. The matrix confirms that rolling factor betas are largely independent of each other and of the price-based signals, justifying inclusion of all features without a-priori dimensionality reduction.

Table 11: Mean feature correlation matrix (2009–2023)

	$\hat{\beta}_M$	$\hat{\beta}_S$	$\hat{\beta}_H$	$\hat{\beta}_R$	$\hat{\beta}_C$	$\hat{\beta}_{Mom}$	Mom	Rev	RVol
$\hat{\beta}_{Mkt}$	1.00	0.12	-0.18	-0.09	-0.21	-0.08	-0.05	0.03	0.31
$\hat{\beta}_{SMB}$	0.12	1.00	-0.07	-0.22	-0.14	-0.04	0.02	-0.01	0.14
$\hat{\beta}_{HML}$	-0.18	-0.07	1.00	0.06	0.41	-0.10	-0.08	0.05	-0.12
$\hat{\beta}_{RMW}$	-0.09	-0.22	0.06	1.00	0.07	0.06	0.04	-0.02	-0.18
$\hat{\beta}_{CMA}$	-0.21	-0.14	0.41	0.07	1.00	-0.06	-0.06	0.04	-0.19
$\hat{\beta}_{Mom}$	-0.08	-0.04	-0.10	0.06	-0.06	1.00	0.42	-0.18	-0.09
Mom ₁₂₋₁	-0.05	0.02	-0.08	0.04	-0.06	0.42	1.00	-0.21	-0.07
Rev ₁	0.03	-0.01	0.05	-0.02	0.04	-0.18	-0.21	1.00	0.04
RVol ₁₂	0.31	0.14	-0.12	-0.18	-0.19	-0.09	-0.07	0.04	1.00

Notes: Mean Pearson correlation across the out-of-sample period (2009–2023), pooled across all asset-month observations. The high correlation between $\hat{\beta}_{Mom}$ and Mom₁₂₋₁ (0.42) and between $\hat{\beta}_{HML}$ and $\hat{\beta}_{CMA}$ (0.41) explains the near-zero SHAP for price momentum and the joint importance of value and investment factor betas.

D Hyperparameter Selection

Table 12 reports the hyperparameter grid searched for each model and the most frequently selected values across the 15 validation folds.

Table 12: Hyperparameter grid and selected values

Model	Parameter	Grid	Most selected
Lasso	λ	$\{10^{-4}, 5 \times 10^{-4}, 10^{-3}, 5 \times 10^{-3}, 0.01, 0.05, 0.1, 0.5\}$	10^{-3}
Random Forest	Max depth	$\{4, 6\}$	4
	Min leaf size	$\{10, 20\}$	20
	Trees	$\{200, 300\}$	200
XGBoost	Learning rate	0.05 (fixed)	—
	Max depth	4 (fixed)	—
	Subsample	0.8 (fixed)	—
	Col sample	0.8 (fixed)	—
	Min child weight	20 (fixed)	—
	Early stopping	30 rounds	87 trees (median)

Notes: XGBoost uses early stopping on validation MSE rather than a grid search. The median number of trees selected across folds is 87, ranging from 42 to 241. Shallow trees (depth 4) are consistently preferred for Random Forest, consistent with the low-dimensional feature space (9 features) requiring limited tree complexity to capture the relevant structure.

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